Saeyoung Kim

Cambridge, MA, USA
+1 (351) 322 5510 • saeyoung_kim@uml.edu • saeyoung-macx-kim
PrimaryAnomaly

Professional Summary

Technical project manager and systems engineer who has led multi-year, cross-border R&D programs in pharmaceutical manufacturing, space hardware, and AI-driven process software. Currently managing the US side of a multi-million-dollar KEIT-funded digital twin program. Previously ran a KRW 300M (~$230K) government-funded CubeSat payload project through 8 years of development. I build the software I manage: full-stack web apps, real-time data pipelines, simulation models, and multi-agent AI systems. Bilingual in English and Korean, comfortable bridging technical teams across time zones and institutions.

Technical Skills

Project Mgmt: WBS Development, Gantt Charts, Critical Path Analysis, Risk Tracking, Milestone Planning, Cross-Team Coordination, Government Project Reporting

Software Dev: Python, C/C++, MATLAB, FastAPI, Next.js, Docker, Git, Linux

PM Tools: Excel WBS/Gantt, PowerPoint Status Decks, Git/GitHub

Data/AI: PCA, PLS, ODE Modeling, TimescaleDB, Telegraf, Real-Time Data Pipelines

Hardware: Circuit Design, PCB Layout (KiCad), Sensor Systems, Embedded Firmware, CAD (Fusion360, SolidWorks)

Project Management Experience

University of Massachusetts Lowell (KEIT-funded) **Lowell, MA, USA**
Project Manager and Technical Lead | *Lyophilization Digital Twin* *2025.07–Present*

- Manage US side of a 5-year KEIT-funded digital twin for pharmaceutical freeze-drying (mRNA-LNP), coordinating a 6-person team with Korean partners
- Maintain a 15-work-package WBS with multi-year Gantt chart across two phases (3-year R&D + 2-year technology transfer)
- Run weekly status meetings with executive summaries, milestone tracking, and risk/blocker identification
- Track procurement of specialized equipment ($194K+ TDLAS, 16-week lead time) and coordinate vendor timelines with project milestones
- Assign task ownership with monthly milestones across parallel workstreams: experimentation, modeling, frontend, backend
- Pivoted from gPROMS to Python when the commercial tool could not support the required data pipeline, keeping the project on schedule
- Built the project's core software: 9-state ODE model, multi-agent AI system (DAEMON), FastAPI + Next.js web app, Telegraf/TimescaleDB pipeline

Korea University (NRF + GPF funded) **Seoul, South Korea**
Project Manager and Technical Lead | *BioCubeSat Biological Payload (CyanOx)* *2017.03–2024.02*

- Led an 8-year, KRW 300M (~$230K) program across two government grants (5-year GPF + 3-year NRF) to build a 3U CubeSat biological payload
- Created WBS across 6 subsystems (Mechanical, Fluidics, Sensors, Control, MCU, Biology) with 12 versioned action plans and Gantt scheduling
- Wrote monthly work plans with per-person task assignments and collected monthly progress reports from each of 4 team members
- Managed the full research budget (personnel, materials, equipment, travel) with NRF/GPF compliance
- Delivered all milestones to the funding agency: kickoff report, 2 annual reports, 1 final report (5 to 8 revision cycles each)
- Ran weekly team meetings covering scope review, Gantt status, development methods, and monthly plan alignment
- Coordinated the design-build-test cycle across 6 subsystems through two full hardware integration rounds
- Published 3 first-author journal papers and delivered 4 conference presentations as project deliverables

Nonprofit Community Management **Seoul, South Korea**

Co-Founder and Lead Organizer | *Effective Altruism South Korea and AMF Korea* *2015–2023*

- Co-founded Effective Altruism South Korea in 2015, built a bilingual community with regular meetups and tracked metrics
- Secured over $100K in grants from the Centre for Effective Altruism for community operations and programs
- Built organizational infrastructure from scratch: OKRs, member pipeline, finances, project tracking, standard processes
- Ran concurrent projects: book club, Korean translation of EA materials, outreach strategy, member surveys
- Helped found AMF Korea as a registered nonprofit in 2023, recruiting promoters and preparing bilingual legal documents

Professional Experience

University of Massachusetts Lowell **Lowell, MA, USA**

Postdoctoral Associate | *Manufacturing Process Monitoring and Test Automation* *2025.07–Present*

- Built Python diagnostic software suite replacing commercial platforms for real-time manufacturing process monitoring
- Developed anomaly detection and quality prediction pipelines across 100+ pharmaceutical production batches
- Designed data-driven test strategies with automated error flagging for multi-day batch manufacturing cycles
- Translated cross-functional stakeholder requirements into standardized test procedures and documentation

Hanwha Systems **Seoul, South Korea**

Systems Integration Engineer | *Test Equipment Design and Satellite Integration* *2024.02–2025.05*

- Designed and deployed EGSE architecture for satellite subsystem factory acceptance testing
- Developed and executed system-level test plans for military SAR satellite qualification across EM, QM, and FM builds
- Delivered turnkey test setups with data acquisition hardware, sensor interfaces, and electrical connections for production
- Performed root cause analysis of electromechanical failures during integration, driving corrective actions to resolution
- Provided design-for-test recommendations based on failure trends and fault isolation during assembly campaigns
- Managed purchasing, integration, and commissioning of functional test systems for factory and field deployment
- Led test equipment reliability improvements by identifying recurring failure modes and implementing fixes
- Wrote and maintained controlled test procedures, fixture documentation, and troubleshooting guides
- Trained engineers and technicians on test equipment operation, debug processes, and troubleshooting methods
- Coordinated across electrical, mechanical, software, and program teams to balance test coverage with schedules

Education

Korea University **Seoul, South Korea**

PhD in Electrical Engineering | *Embedded Sensor Systems and Hardware Test Development* *2017.03–2024.02*

- Secured and managed KRW 300M (~$230K) in research grants for CubeSat biological payload development
- Built custom test fixtures and data acquisition setups using oscilloscopes, DMMs, function generators, and power supplies
- Designed multi-modal sensor suite through full cycle: schematics, PCB layout, fabrication, assembly, and qualification
- Developed MCU firmware in C/C++ for real-time multi-channel data acquisition over UART, SPI, and I2C
- Performed iterative environmental and reliability testing to identify failure modes and optimize fabrication parameters
- Diagnosed hardware/firmware interaction failures and implemented corrective actions through multiple revision cycles
- Published 3 first-author and 10+ co-authored journal papers on sensors, test systems, and space payload hardware

Konkuk University **Seoul, South Korea**

BSc in Mechanical Engineering *2013.03–2017.02*

- Conducted research in electrochemical sensor microfabrication, cleanroom processing, and hardware prototyping
- Gained hands-on experience in CAD modeling, mechanical assembly, PCB prototyping, and precision measurement

Publications

Journal Publications

- **Saeyoung Kim**, Jaewon Lee, Kimia Babaei, Seonkgyu Yoon, "Digital Twins in Lyophilization: Advances in Pharmaceutical Process Development and Optimization," *Biotechnology Advances*, 2026. (Manuscript in Preparation, Target Submission: Q1 2026)
- Jinhwee Kil, **Saeyoung Kim**, James Jungho Pak, "Investigation of thermal management strategies for biological CubeSat payloads," *Acta Astronautica*, 2024, Art. no. 228.
- **Kim, S.**; Park, S.; Pak, J.J., "Multi-Modal Multi-Array Electrochemical and Optical Sensor Suite for a Biological CubeSat Payload," *MDPI Sensors*, 2024, 24, Art. no. 265.
- Ji-Hoon Han, Sang Hyun Park, **Saeyoung Kim**, James Jungho Pak, "A Performance Improvement of Enzyme-Based Electrochemical Lactate Sensor Fabricated by Electroplating Novel PdCu Mediator on a Laser Induced Graphene Electrode," *Bioelectrochemistry*, 2022, 148, 108259.
- **Saeyoung Kim**, Ji-Hoon Han, Beelee Chua, James Jungho Pak, "A pH Sensing Pipette for Cross-Contamination Prevention in Industrial Fermentation," *IEEE Trans. Industrial Electronics*, 2022, 69(7), 7461-7469.
- S.D. Raut, N.M. Shinde, B.G. Ghule, **S. Kim**, J.J. Pak, Q. Xia, R.S. Mane, "Room-Temperature Solution Processed Sharp-Edged Nanoshapes of," *Chemical Engineering Journal*, 2022, 433(2), 133627.
- S.E. Jeong, **S. Kim**, J.H. Han, J. Jungho Pak, "Simple Laser-Induced Graphene Fiber Electrode Fabrication for High-Performance Heavy-Metal Sensing," *Microchemical Journal*, 2022, 172, 106950.
- Sanghoon Cho, Jungmo Jung, **Saeyoung Kim**, James Pak, "Conduction Mechanism and Synaptic Behaviour of Interfacial Switching AlOσ-based RRAM," *Semiconductor Science and Technology*, 2020, 35(8), 085006.
- Ji-Hoon Han, **Saeyoung Kim**, Jaesung Choi, Sora Kang, Youngmi Kim Pak, James Jungho Pak, "Development of Multi-Well-Based Electrochemical Dissolved Oxygen Sensor Array," *Sensors and Actuators B: Chemical*, 2020, 306, 127465.

Conference Presentations

- *Thermal Management System for a Fluidic Card in a Small Satellite Biological Payload*, Korean Institute of Electrical Engineers (KIEE) Autumn Conference, 2022.
- *A Pipette with an Integrated pH Sensor for Liquid Sampling Applications*, Korean Institute of Electrical Engineers (KIEE) Summer Conference, 2019.
- *IoT System for Monitoring pH, Dissolved Oxygen, Temperature, and Electrical Conductivity in Liquids*, Korean Institute of Electrical Engineers (KIEE) Summer Conference, 2018.
- *Development Trends of Non-Invasive Glucose Sensor Technology*, Korean Institute of Electrical Engineers (KIEE) Summer Conference, 2017.

Seminar Presentations

- *Bioprocessing in Space: Challenges and Opportunities for Pharmaceutical Manufacturing Beyond Earth*, Lowell Center for Space Science and Technology (LoCSST), UMass Lowell, 2025.

References available upon request.